# Starting and stopping the service without losing a request

Shutdown is the delicate half: it must stop workers blocked on an empty queue without abandoning work already in flight

### 1. Start

Allocate the shards, spawn one worker thread per shard, and hand each worker exclusive ownership of its store. Nothing is shared, so nothing needs locking.

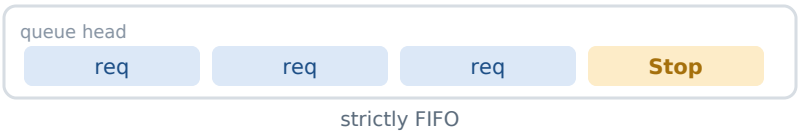
### 2. Run

A request is moved into the queue rather than copied, carrying its completion handle with it, so the producer and the worker never both hold a usable reference to it.

### 3. Stop

Push one Stop sentinel onto each shard's queue, then join the workers, so shutdown returns only once every shard has actually stopped.

## Why an in-band sentinel needs no drain API




The worker observes Stop only after the three requests ahead of it have run and completed, so nothing is abandoned.

### Tied to object destruction

The C++ destructor calls shutdown() and Rust's Drop does the same, so a service that goes out of scope cannot leave detached threads behind. A running flag makes the call idempotent on both sides.

### The one piece of shared mutable state

An atomic activity counter, held by reference in C++ and as a cloned Arc<AtomicU64> in Rust, incremented with relaxed ordering because nothing branches on its value. Write-only bookkeeping.

No separate closed flag, timeout, or drain API is required, and no completion handle is left unfulfilled.